



## 4. MASS SPECTROMETRY

Instrumentation in Chemical Analysis (University of Alberta)

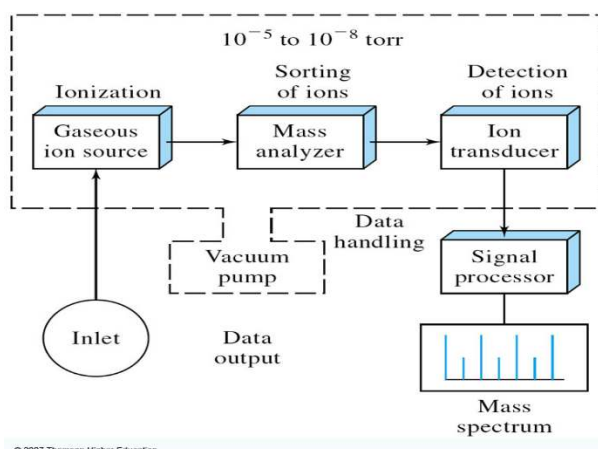
# MASS SPECTROMETRY

## I. What is Mass Spec?

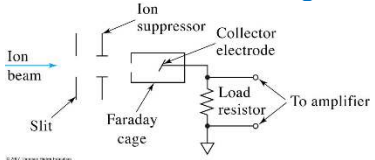
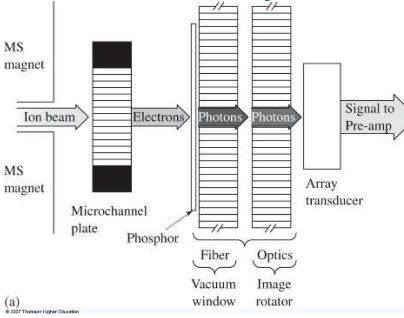
1. **Definition:** To separate a mixture of ions based on their masses and their charges. Output are typically a plot of intensity vs mass-to-charge ratio ( $m/z$ ).
2. **Nominal mass:** of an ion is calculated by simply adding the integer (whole number) masses of the lightest isotopes of all elements contributing to the molecule.  
**Exact mass:** are masses reported in the literature are EXACT masses.  
**Isotope peak:** Ionic fragments of the same chemical formula are usually represented by multiple adjacent peaks attributable to ions of different isotopic compositions. The intensity of each peak (from  $M$  to  $M+5$ ) is proportional to the probability of occurrence of each mass.  
**Mass-to-charge ratio:** in stage 3, the two factors (1.mass and 2.charge of the ions) are combined into the **Mass-to-charge ratio ( $m/z$ )**. For example: if an ion had a mass of 28 and a charge of +1, its  $m/z=28$ ; or if the charge is +2, its  $m/z=14$ .
3. **Mass resolving power:** measures of the ability to distinguish two peaks of slightly different mass-to-charge ratios  $\Delta m = m_2 - m_1$ . Usually at 50% of the peak height. Basically, mass resolving power is a measure of how "fat" a peak is, the bigger the number the skinnier the peak.  
**Mass resolution:** refers to the ability of separating two narrow mass spectral peaks, is calculated as  $m_2/(m_2 - m_1)$   $\Delta m$  is the mass difference between two adjacent peaks that are just resolved and  $m$  is the nominal mass of the first peak.  
**Example:** What resolution needed to separate  $C_2H_4^+$  (28.0313) and  $CH_2N^+$  (28.018724)?  $R=2226.5$

## II. How does a Mass Spectrometry work?

4. **Outline of what happens:** There are 4 stages of the analytical process in a mass spectrometer  
**Stage 1: Ionisation** (The atom is ionised by knocking one or more electrons off to give a positive ion. Most MS work with positive ions).  
**Stage 2: Acceleration** (The ions are accelerated so that they all have the same kinetic energy)  
**Stage 3: Deflection** (The ions are then deflected by a magnetic field according to their mass  $\rightarrow$  lighter, more deflected. The amount of deflection also depends on how many electrons were knocked off  $\rightarrow$  more ion is charged, the more it gets deflected)  
**Stage 4: Detection** (The beam of ions passing through the machine is detected electrically)
5. **The diagram of a mass spec:**



6. **Detectors for Mass Spec:**

| FARADAY CUP (single channel)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ELECTRO MULTIPLIERS (single)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ARRAY TRANSDUCERS                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a metal (conductive) cup designed to catch charged particles in vacuum. The resulting current can be measured and used to determine the number of ions or electrons hitting the cup.</p>  <ul style="list-style-type: none"> <li>- Independent of mass, energy and chemical nature.</li> <li>- Simple, rugged and inexpensive.</li> <li>- High-impedance amplifier: slow scanning.</li> <li>- Low sensitivity</li> </ul> | <p>High sensitivity.<br/>Fast response.<br/>Required high kinetic energy beam</p> <p><b>Discrete Dynode:</b> (<math>10^7</math> e/ion)<br/>a photo-emissive surface is followed by an electron multiplier with several sequential multiplying electrodes called dynodes. Because these electrodes are separate from each other, this is called a "discrete-dynode" multiplier.</p> <p><b>Continuous Dynode:</b> (<math>10^5</math>-<math>10^8</math> e/ion)<br/>Ion-to-electron detector in which the <u>ion</u> strikes the inner surface of a continuous tube device and induces the production of secondary electrons that, in turn, impinge on the inner surfaces of the device to produce more secondary electrons.</p> | <p>Microchannel plate—an array detector for ions.</p> <ul style="list-style-type: none"> <li>- Electro-optical ion detector (electrons and light)</li> <li>- Analogous to optical detectors, detector arrays allow multiple ion trajectories (different ions) to be monitored simultaneously.</li> </ul>  <p>(a)</p> |

### III. MASS ANALYZER:

Sort ions based upon  $m/z$  ratio. Exert forces upon ions in order to differentiate  $m/z$  ratios.

Ions can be manipulated by electric and magnetic fields. Ions that traverse a magnetic field have a bent trajectory.

#### 1. Magnetic sector:

- Ions are injected into a circular path (electro)magnet and only those of a certain  $m/z$  ratio can balance the centripetal (FC) and magnetic forces (FM) in successfully traverse the path; others hit the sides.
- Disadvantage: Not all ions of same  $m/z$  have identical kinetic energy
- Kinetic energy of ions accelerated out of source:

$$KE = zeV = \frac{mv^2}{2} \quad \text{z: charge; e: charge of e; V: voltage; v: velocity}$$

- Force from magnet balances with centripetal force:

$$\frac{m}{z} = \frac{B^2 r^2 e}{2V} \quad \text{r: radius; B: field strength}$$

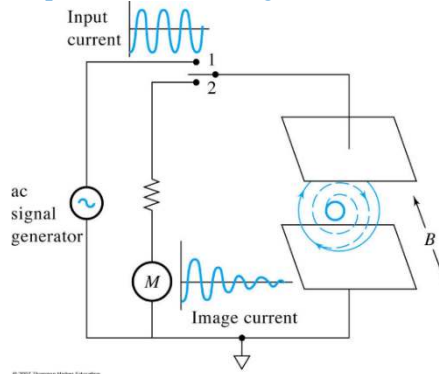
#### 2. Double-Focusing MS:

- Ions are first passed through electrostatic energy filter to isolate a single kinetic energy. Ions formed in source have range of translational energies.
- Double-focusing corrects for directional and energetic variations (electrostatic sector and magnetic sector)
- High resolution:  $> 10^5$

#### 3. Time-of-Flight (TOF) MS:

- Ions are generated in pulses and accelerated (also in pulses, but with a lag) in a  $10^3$ - $10^4$  V field into a field-free drift tube. Accelerate ions into fields-free region (all ions have same KE).
- Can optimize for speed or resolution/ Very simple and rugged/ Theoretically unlimited mass range/ Require very fast electronics: range of flight times is a few microseconds.

- Ions separate based on flight times:



$$t_F = \frac{L}{v} = L \sqrt{\frac{m}{2zeV}}$$

$$\frac{m}{z} = 2eV \left( \frac{t_F}{L} \right)^2$$

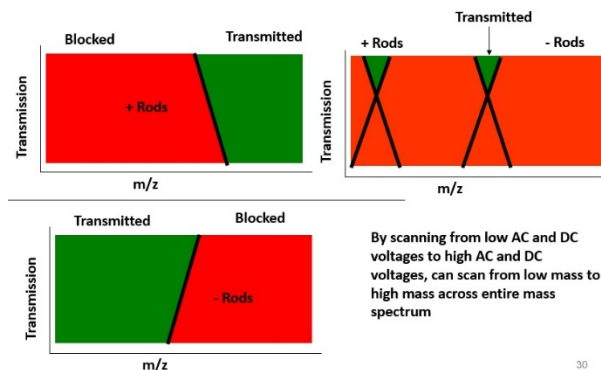
Resolution better than, but sensitivity (quantitation) not as good as quadrupoles.

Advantages: simplicity, ruggedness, virtually unlimited range, and fast data acquisition.

#### 4. Quadrupole Mass Spectrometer:

- Mass isolation is achieved with alternating applied electric fields. Only ions having a specific  $m/z$  ratio are confined within and traverse the entire quadrupole without becoming unstable and colliding with a rod.
- Positive rods:** Ions with a high  $m/z$  have more inertia and respond slowly to influence of AC voltage  $\Rightarrow$  they do not stray from centre. Lighter ions respond quickly to influence of AC voltage and are ejected from quadrupole. Positive rods form a “High-pass filter” (high mass pass through).
- Negative rod:** Ions with a high  $m/z$  have more inertia and respond slowly to influence of AC voltage  $\Rightarrow$  they are drawn to rods. Lighter ions respond quickly to influence of AC voltage and pushed back to centre. Positive rods form a “Low-pass filter” (high mass pass through).

Together, Quadrupole = mass filter



#### 5. FT-resonance):

#### ICR (Fourier-transform ion cyclotron

- FT-ICR is a type of mass spectrometry particularly suited to identifying heavy molecules. is based on the circular movement of charged particles in a strong magnetic field (cyclotron movement). The cyclotron frequency depends directly on the mass-to-charge ratio of the ions.

$$\omega_c = \frac{v}{r} = \frac{zeB}{m}$$

- Ions absorb energy from electric field with resonant frequency, increase their velocity and radius, then start to circulate in a coherent manner. Pulse ions then measure image current, magnitude of current increases with number of ions.
- Very high resolution:  $>10^6$
- 1. Apply input current  $\rightarrow$  AC signal flip switch to 2- $\rightarrow$  measure at image current.

#### IV. ICP MS (Inductively coupled plasma mass spectrometry):

Multi-elemental analytical method that has a very low limits of detection.

Advantage: Better detection limit than OES and ETAAS, Speed for multielement analysis.

### 1. How does it work?

- Samples are introduced into an argon plasma as aerosol droplets. The plasma dries the aerosol, dissociates the molecules, and then removes an electron from the components, thereby forming singly-charged ions, which are directed into the mass spectrometer. Upon exiting the mass spectrometer, ions strike the first dynode of an electron multiplier, which serves as a detector.
- LA-ICP-MS begins with a laser beam focused on the sample surface to generate fine particles. Benefits of this method: *can perform ultra-highly sensitive chemical analysis down to ppb level — without any sample preparation, Samples can be both conducting or non-conducting, the sample mass size required is sub-microscale*
- ICP Torch: generates the argon plasma, which serves as the ion source of the ICP-MS.
- ICP-EOS Torch: Using the electromagnetic field created in the torch tube by the high frequency current, argon gas is ionized and plasma is generated.

### 2. Spectroscopic Interferences: Can cause a decrease in signal tho not always.

- Isobaric: species with small difference in m/z.
- Polyatomic ions: caused by reaction of plasma atmosphere.
- Doubly-charged ions
- Oxides and hydroxides of analyte form in the plasma.

### 3. Minimize Interferences:

- Diluting samples.
- Changing sample introduction
- Separating offending species.
- Appropriate internal standards.

### 4. Quantitative ICP-MS:

- **External calibration curve** is typically used (total dissolved solids), Internal standard (In and Rh)
- **Isotope dilution** is the most accurate: intensity for two isotopes of analyte, spike with solution enriched in one isotope, measure the change in the ratio of the two signals, slow technique.

$$C_X = C_S \left( \frac{{}^1A_S - R_{1/2} \times {}^2A_S}{{}^1A_X - R_{1/2} \times {}^2A_X} \right)$$

$C_X$  = Concentration of unknown

$C_S$  = Concentration of spike

${}^1A_X$  = Fraction of isotope 1 in X

${}^2A_X$  = Fraction of isotope 2 in X

${}^1A_S$  = Fraction of isotope 1 in S

${}^2A_S$  = Fraction of isotope 2 in S

$R_{1/2}$  = Ratio of isotopes in spiked sample

- **Isotope Ratio methods:** ageing of artifacts, isotopically enriched medical tracers, biogenic vs. anthropogenic sources of chemicals.

## **V. MOLECULAR MASS SPEC:**

A type of spectrometry where the sample is converted into gaseous molecular ions, which are characterised by their mass-to-charge ratio and relative abundance.

Appearance of mass spectra depends on ionization method:

1. Gas phase – molecules are vaporized then ionized
2. Desorption – solid/ liquid directly into gaseous ions

### 1. PEAK:

a) **Base Peak**: The most intense (tallest) peak in a mass spectrum, because it represents the commonest fragment ion to be formed - either because there are several ways in which it could be produced during fragmentation of the parent ion, or because it is a particularly stable ion.

b) **Molecular Peak**: an electron is removed from the sample molecule; the resultant radical cation is called the molecular ion (symbols:  $M^+$ ,  $M^+$ ), the peak representing the molecular ion is called the molecular ion peak.

## 2. FRAGMENTATION:

Definition: the dissociation of energetically unstable molecular ions formed from passing the molecules in the ionization chamber of a mass spectrometer. The fragments of a molecule cause a pattern in the mass spectrum used to determine structural information of the molecule. *Fragmentation can be an advantage or disadvantage; can add certainty or uncertainty.*

## 3. IONIZATION:

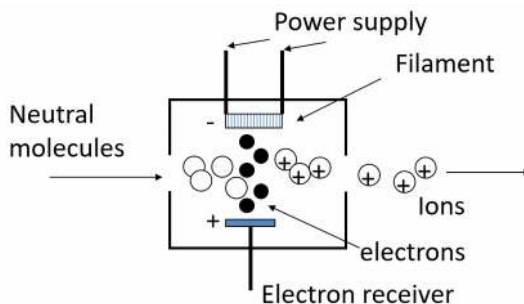
Definition: The production of gas phase ions suitable for resolution in the mass analyser or mass filter. Ionization occurs in the ion source. There are several methods of ionization:

### a) Hard Ionization:

- Definition: HI techniques are processes which impart high quantities of residual energy in the subject molecule invoking large degrees of fragmentation. Resultant ions tend to have  $m/z$  lower than the molecular mass.
- Example of hard ionization: EI, CI.

#### i. Electron ionization:

- Energetic electron beam causes molecules to lose an electron (become ionized) due to electrostatic repulsion.
- EI Spectra: electrons require  $>50V$  to be effective (Light – little change in translational energy/ Molecules in highly excited Vibrational and Rotational states). Extensive fragmentation (Generate product ions/ Relatively weak  $M^+$ / very useful for structure elucidation).
- **Advantages**: Sensitive (produce high ion currents), Extensive fragmentation (good for identification)
- **Disadvantages**: Need to volatilize sample ( if thermal degradation of analyte occurs, limits technique to mass  $< \sim 10^3$  Da)
- Heat source generates molecular vapour. e- beam from filament strikes molecules. Generates singly-charged + molecular ions.



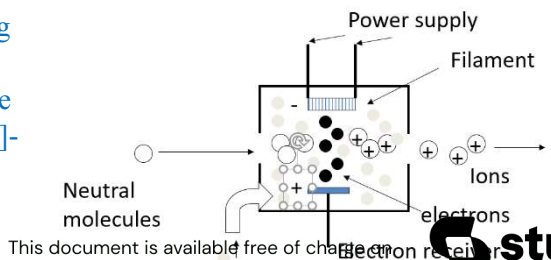
#### ii. Chemical

#### Ionization:

- Chemical ionization technique uses virtually the same ion source device as in electron impact, except, CI uses tight ion source, and reagent gas. Sample ions are formed by reaction with ions of reagent gas, can do + or – ionization.
- There are two types of reagent gas:

1 - PCI (Positive  
 $C_4H_{10}$ ,  $NH_3$ . Using  
 transfer  $(M-1)^+$ :  
 2 – NCI (Negative  
 forms  $OH^-$ :  $[G-H]^-$ -

chemical ionization) like  $CH_4$ ,  $H_2$ , i-  
 proton transfer  $(M+1)^+$  and hydride  
 $GH^+ + M \rightarrow MH^+ + G$ .  
 chemical ionization) like  $N_2O + H_2O$  to  
 $+ M \rightarrow [M-H]^- + G$ .

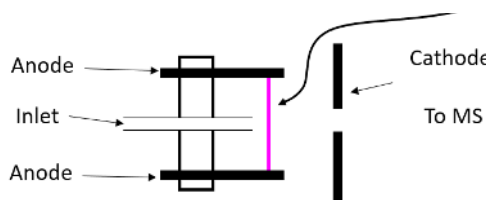


### b) Soft Ionization:

- Definition: the processes which impart little residual energy onto the subject molecule and as such result in little fragmentation.
- Desorption method: formation of gas phase ions from a solid or liquid sample surface upon activation by heat, strong electric field, particles, droplets, or photon bombardment. Simpler spectra, tho exact mechanisms unknown.
- Example of Soft ionization: FI, MALDI, ESI, DESI.

#### i. Field Ionization:

- Multi-tipped electrode is coated with sample solution. Electrode is inserted into compartment and a high voltage ( $10^8\text{V/m}$ ) is applied. The sample ionizes and goes into the gas phase by electrostatic repulsion. Produce very little fragmentation.
- Not a very sensitive method: 1/10 that of EI



#### ii. Matrix (MALDI):

#### Assisted Laser Desorption Ionization

- Sample is dispersed in matrix (usually a crystalline solid) and exposed to laser beam. Matrix material absorbs radiation and “explodes” producing desorbed and ionized analyte molecules.
- Ionization process: deposit dilute sample on solid probe. Place in vacuum chamber. Focus laser onto sample. Matrix absorbs radiation is volatilized.
- Use extensively in bio field (large organic compounds).
- Common MALDI matrices: nitropyridine, Benzoic acid, etc.

#### iii. Electrospray Ionization (ESI):

- Ionization process: Sample (in solution) passed through needle biased. Charged droplets pass through capillary where solvent evaporates, and molecules begin to inherit the droplets charge. Droplets shrink. Explosion occurs and the droplet bursts into smaller droplets.
- The most important technique for biomolecules.
- No fragmentation – disadvantage in some applications. Use atmospheric temperatures and pressure. Easily adapt to Liquid Chromatography system.

#### iv. Desorption Electrospray ionization (DESI):

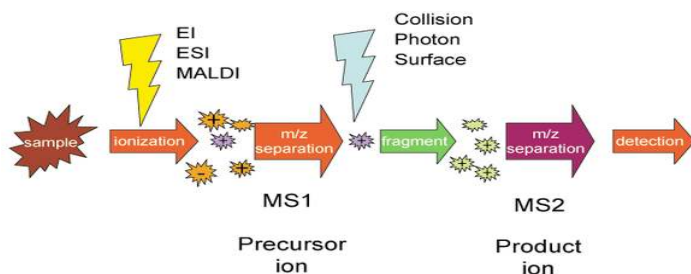
- A fine spray of charged droplets hits the surface of interest, from which it picks up small organic molecules and large biomolecules, ionizes them, and delivers them—as desolvated ions—into the mass spectrometer.

### 4. TANDEM MASS SPECTROMETRY (MS/MS):

- **DEFINITION**: involves multiple steps of mass spectrometry selection, with some form of fragmentation occurring in between the stages. A pre-selected ion is fragmented, and the pieces analyzed again by mass spectrometry.



- **Tandem in space:** In tandem mass spectrometry *in space*, the separation elements are physically separated and distinct.
- **Tandem in time:** In tandem mass spectrometry *in time*, the separation is accomplished with ions trapped in the same place, with multiple separation steps taking place over time.
- **PROCESS:** ions are formed in the ion source and separated by mass-to-charge ratio in the first stage of mass spectrometry (MS1). Ions of a particular mass-to-charge ratio (precursor ions) are selected and fragment ions (product ions) are created by collision-induced dissociation, ion-molecule reaction, etc. The resulting ions are then separated and detected in a second stage of mass spectrometry (MS2).



- Multiple selected reaction monitoring: q1 selects the ion of interest, q2 collision gas causes fragmentation of ion of interest, q3 monitors only the mass for a particular daughter ion. Analyte must have correct mass for Q1 and must also generate ion with correct mass for Q3.
- Precursor Ion Scan: analytes of interest all give specific m/z. set q3 to a specific m/z, scan q1. If compound ionizes to  $M^+$  and  $M^+$  gives out that specific m/z as product  $\rightarrow$  get signal.
- In a product ion scan: a precursor ion is selected in the first stage, allowed to fragment and then all resultant masses are scanned in the second mass analyzer and detected in the detector that is positioned after the second mass analyzer.
- INCREDIBLY SELECTIVE TECHNIQUE!!!!

### 5. ION EXCHANGE CHROMATOGRAPHY:

- Generally used for separation of small charged species. Uses columns packed with ion exchange resin.
  - Ion exchange selectivity: Resin excludes ions with same charge. Ions with opposite charge to resin compete with other for binding sites. Selectivity increases with cross-linking: smaller pore size  $\rightarrow$  ions with large hydrated radius have restricted access.
  - Resins favour ions with higher charge, smaller hydrated radius, increased polarizability (increase going the PT).
  - Consider 2 anions A<sup>-</sup> and B<sup>-</sup> in a sample that we want to separate. Eluate with a gradient of anion C<sup>-</sup> that binds more weakly than A<sup>-</sup> or B<sup>-</sup>.
  - $R^+ B^- + C^- \rightleftharpoons R^+ C^- + B^-$
- $R + C \rightleftharpoons$   
 $B \rightleftharpoons$   
 $R + B \rightleftharpoons$   
 $C \rightleftharpoons$   
 $Q \rightleftharpoons$